

UNGSKAR, Sweden

WMO: 026660

Lat: 56.041N

Lon: 15.804E

Elev: 3

StdP: 101.29

Time Zone: 1.00 (EUC)

Period: 86-95

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-10.1	-6.7	-13.2	1.2	-8.0	-10.6	1.5	-5.5	20.7	5.0	19.0	1.8	4.9	320	0.852

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	3.8	22.4	18.7	21.1	17.8	19.9	17.0	19.3	21.6	18.3	20.5	17.4	19.4	5.2	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.4	13.3	20.7	17.4	12.5	19.7	16.4	11.7	18.8	55.1	21.5	51.8	20.4	48.9	19.6	23.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.9	16.1	14.7	-9.9	23.4	5.1	2.9	-13.6	25.5	-16.6	27.1	-19.5	28.7	-23.2	30.8		
(5)			-10.6	19.5	4.8	2.2	-14.1	21.1	-16.9	22.3	-19.6	23.5	-23.1	25.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	7.8	1.4	0.8	2.0	4.9	9.6	13.7	16.4	15.9	12.6	8.8	4.8	2.7	(6)
(7)		DBStd	6.22	3.90	3.78	3.21	2.42	2.40	2.55	2.48	2.54	2.14	2.67	2.75	2.67	(7)
(8)		HDD10.0	1404	266	259	249	154	37	2	0	0	4	53	156	225	(8)
(9)		HDD18.3	3851	524	492	507	403	271	139	72	86	171	296	406	483	(9)
(10)		CDD10.0	617	0	0	0	1	25	113	197	182	83	16	0	0	(10)
(11)		CDD18.3	22	0	0	0	0	0	1	11	10	0	0	0	0	(11)
(12)		CDH23.3	17	0	0	0	0	0	1	13	3	0	0	0	0	(12)
(13)		CDH26.7	1	0	0	0	0	0	0	1	0	0	0	0	0	(13)
(14)	Wind	WSAvg	7.6	9.4	9.0	8.3	7.3	6.6	6.2	6.0	7.3	7.8	7.8	8.9	(14)	
(15)	Precipitation	PrecAvg	628	47	43	30	31	48	51	65	61	59	65	62	(15)	
(16)		PrecMax	793	98	80	58	73	127	130	200	150	183	152	164	144	(16)
(17)		PrecMin	467	0	12	1	3	8	0	6	5	10	17	10	28	(17)
(18)		PrecStd	86	25	21	15	19	24	26	44	37	47	32	35	27	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.0	7.7	7.9	11.9	17.6	21.2	25.0	23.7	18.5	14.1	10.3	8.3	(19)
(20)			MCWB	5.9	6.4	5.7	8.6	13.6	17.0	19.1	18.9	16.0	12.4	8.9	7.1	(20)
(21)		2%	DB	6.3	6.5	6.7	10.4	15.9	19.8	22.9	22.3	17.0	13.3	9.7	7.6	(21)
(22)			MCWB	5.3	5.1	5.1	8.0	13.0	16.3	19.3	18.5	14.7	11.8	8.4	6.7	(22)
(23)		5%	DB	5.8	5.7	6.0	9.4	14.6	18.7	21.5	21.2	16.2	12.7	9.1	6.9	(23)
(24)			MCWB	4.8	4.5	4.6	7.4	12.2	15.5	18.2	17.9	14.2	11.4	8.0	5.9	(24)
(25)		10%	DB	5.3	4.9	5.2	8.3	13.5	17.8	20.2	19.9	15.6	12.2	8.3	6.2	(25)
(26)			MCWB	4.3	3.8	4.0	6.5	10.9	15.0	17.3	17.3	13.7	11.1	7.1	5.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.2	6.6	6.2	9.4	14.7	18.0	21.1	20.3	16.9	12.9	9.4	7.6	(27)
(28)			MCDWB	6.5	7.4	7.3	11.0	16.5	20.4	23.3	22.4	17.8	13.6	9.9	7.9	(28)
(29)		2%	WB	5.6	5.2	5.4	8.4	13.6	17.1	19.7	19.2	15.6	12.3	8.7	6.7	(29)
(30)			MCDWB	6.1	6.1	6.4	9.9	15.5	19.0	22.2	21.3	16.4	12.9	9.4	7.4	(30)
(31)		5%	WB	5.0	4.7	4.8	7.5	12.5	16.3	18.6	18.5	14.8	11.8	8.1	6.1	(31)
(32)			MCDWB	5.7	5.5	5.7	9.1	14.4	18.4	20.8	20.5	15.7	12.5	8.9	6.7	(32)
(33)		10%	WB	4.4	4.0	4.2	6.6	11.3	15.3	17.7	17.4	14.1	11.3	7.4	5.4	(33)
(34)			MCDWB	5.2	4.7	5.1	8.0	13.3	17.4	19.7	19.6	15.3	12.1	8.1	6.1	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	2.6	2.9	2.6	3.4	3.8	3.8	3.8	3.3	3.1	3.2	3.0	(35)	
(36)			MCDBR	1.8	2.1	2.7	4.4	5.3	5.1	5.4	4.8	4.1	3.1	3.4	2.7	(36)
(37)			MCWBR	2.2	1.8	2.3	3.0	3.3	3.3	3.1	2.7	3.0	2.7	3.3	2.8	(37)
(38)		5% WB	MCDWR	1.8	1.9	2.3	4.0	4.7	4.5	4.5	3.1	2.7	2.8	2.5	(38)	
(39)			MCWBR	2.2	1.9	2.3	2.9	3.4	3.1	3.1	2.9	2.7	2.6	3.1	2.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.315	0.340	0.358	0.391	0.388	0.380	0.405	0.405	0.374	0.361	0.343	0.308	(40)	
(41)		taud	2.290	2.311	2.322	2.303	2.379	2.437	2.385	2.385	2.444	2.389	2.308	2.258	(41)	
(42)		Ebn at Noon	584	716	801	826	853	865	831	806	781	685	539	489	(42)	
(43)		Edh at Noon	54	76	95	112	110	106	109	103	85	71	53	44	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.41	1.02	2.47	4.25	5.62	6.02	5.46	4.42	3.06	1.44	0.55	0.29	(44)	
(45)		RadStd	0.07	0.19	0.23	0.48	0.53	0.38	0.60	0.37	0.34	0.24	0.10	0.03	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (37 neighbors)	+0.40	N/A	N/A	N/A	N/A	N/A	N/A	-134	N/A	N/A

Nomenclature: See separate page